

Model Question paper
(Theory – End Exam: Theory)
1046235110– System Administration
October 2025
Time – Two hours
(Maximum Marks: 60)

- N.B.**
1. Answer all questions under Part-A. Each question carries 1 mark.
 2. Answer any 5 questions under Part-B. Each question carries 2 marks.
 3. Answer any 3 questions under Part-C. Each question carries 10 marks.

PART – A

- 1) What is typically installed after the operating system to ensure hardware functions properly?
 - a) Media player
 - b) Drivers
 - c) Office software
 - d) Web browser extensions
- 2) What does ethics in system administration primarily involve?
 - a) Advertising company products
 - b) Maintaining professional appearance
 - c) Using admin privileges responsibly and protecting data privacy
 - d) Increasing system speed at all costs
- 3) Which operating system is mainly used in embedded systems?
 - a) Real-time operating system
 - b) Time-sharing operating system
 - c) Distributed operating system
 - d) Multi-user operating system

- 4) What is one key feature introduced in Windows 10?
- a) Removal of the Start Menu
 - b) Internet Explorer as the default browser
 - c) Virtual desktops and Cortana
 - d) Lack of update support
- 5) Which type of files are targeted during a Disk Cleanup?
- a) System logs, media files, archived documents
 - b) Temporary files, recycle bin contents, thumbnails
 - c) Antivirus files, active processes, kernel files
 - d) Application data, registry keys, boot files
- 6) An application is not connecting to the internet. You suspect the Windows Firewall is blocking it. What should you do to test this?
- a) Restart the application
 - b) Temporarily disable Windows Defender
 - c) Check the Inbound and Outbound rules in Windows Firewall
 - d) Clear the app cache
- 7) Which Windows tool shows detailed system performance metrics like CPU, memory, disk, and network usage?
- a) Resource Monitor
 - b) Command Prompt
 - c) Quick Assist
 - d) Run
- 8) An application on your PC is freezing frequently. What tool can you use to check which processes are using the most CPU or memory?
- a) Resource Monitor or Task Manager
 - b) Event Viewer
 - c) Disk Cleanup
 - d) Windows Backup utility

- 9) What does the command `pwd` display?
- a) The password file
 - b) The present working directory
 - c) The list of users
 - d) The previous directory
- 10) You are using Ubuntu and need to install the package `git`. Which command should you apply?
- a) `yum install git`
 - b) `apt install git`
 - c) `dnf get git`
 - d) `rpm -i git`
- 11) The Linux kernel is written in which programming language?
- a) C
 - b) C++
 - c) Java
 - d) Python
- 12) Which symbol represents the current directory in Linux?
- a) `~`
 - b) `..`
 - c) `/`
 - d) `.`
- 13) What does it indicate if the command `ping 8.8.8.8` works but `ping www.google.com` fails?
- a) The network interface is disabled
 - b) The DNS service is not functioning properly
 - c) The system is offline
 - d) The router is not connected

14)The command `free -m` is used to display:

- a) Disk usage in megabytes
- b) Memory usage in megabytes
- c) CPU utilization percentage
- d) Network usage in MB

15)Which configuration file is used to set SSH server settings in Linux?

- a) `/etc/ssh/sshd_config`
- b) `/etc/ssh/ssh_config`
- c) `/etc/services`
- d) `/etc/network/interfaces`

16)In Linux file permissions, what does the permission value “755” indicate?

- a) Read, write, and execute for owner; no access for others
- b) Read, write, and execute for everyone
- c) Full access for owner, read and execute for group and others
- d) Read-only for all users

17)When a DHCP client sends a DHCP Discover message, it is attempting to:

- a) Confirm an assigned IP address
- b) Request a lease renewal
- c) Locate a DHCP server in the network
- d) Release its IP address

18)Understanding DNS hierarchy, the Top-Level Domain (TLD) in `www.example.com` is:

- a) `www`
- b) `example`
- c) `com`
- d) `dns`

19) Which topology provides the best fault tolerance?

- a) Bus
- b) Ring
- c) Mesh
- d) Star

20) Which protocol is commonly used for VPN encryption?

- a) PPTP
- b) HTTP
- c) SMTP
- d) FTP

PART – B

21) Mention three reasons why system administration is important.

22) Explain the differences between Windows Backup utility and System Restore in terms of data protection and recovery scope.

23) A recently installed application is causing errors. Describe the steps you would take to troubleshoot and fix the issue.

24) Compare file manipulation commands such as cp, mv, and rm in terms of functionality.

25) Apply a command to list all installed packages on a YUM-based system.

26) How does SSH configuration enhance system security during remote access?

27) Explain how to start, stop, and check the status of a service using systemctl.

28) Explain the working process of DHCP.

PART – C

- 29) Explain the different types of operating systems
- 30) Explain the installation and configuration of Antivirus software.
- 31) Describe the file and directory manipulation commands in linux.
- 32) Examine how system resource monitoring tools help in diagnosing performance bottlenecks.
- 33) Explain the concept and working of Intrusion Detection Systems (IDS) .
